

Foundations of Robotics

Homework 4

1. Which of the following matrices are transformation matrices? Explain why (not).

$$T_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_2 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

$$T_3 = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_4 = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & -1 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2. (Exercise 3.15(c)) Please solve the exercise introduced as follows.

(c) Find T_{ca} for the following transformations:

$$T_{ab} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & -1 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad T_{cb} = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & 1 & 0 & 1 \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

3. (Exercise 3.16) Please solve the exercise introduced as follows.

Exercise 3.16 In terms of the $\hat{x}_s, \hat{y}_s, \hat{z}_s$ coordinates of a fixed space frame $\{s\}$, frame $\{a\}$ has its $\hat{x}_a$ -axis pointing in the direction $(0, 0, 1)$ and its $\hat{y}_a$ -axis pointing in the direction $(-1, 0, 0)$, and frame $\{b\}$ has its $\hat{x}_b$ -axis pointing in the direction $(1, 0, 0)$ and its $\hat{y}_b$ -axis pointing in the direction $(0, 0, -1)$. The origin of $\{a\}$ is at $(3, 0, 0)$ in $\{s\}$ and the origin of $\{b\}$ is at $(0, 2, 0)$ in $\{s\}$.

- (a) Draw by hand a diagram showing $\{a\}$ and $\{b\}$ relative to $\{s\}$.
- (b) Write down the rotation matrices R_{sa} and R_{sb} and the transformation matrices T_{sa} and T_{sb} .
- (c) Given T_{sb} , how do you calculate T_{sb}^{-1} without using a matrix inverse? Write T_{sb}^{-1} and verify its correctness using your drawing.
- (d) Given T_{sa} and T_{sb} , how do you calculate T_{ab} (again without using matrix inverses)? Compute the answer and verify its correctness using your drawing.
- (e) Let $T = T_{sb}$ be considered as a transformation operator consisting of a rotation about $\hat{x}$ by -90° and a translation along $\hat{y}$ by 2 units. Calculate $T_1 = T_{sa}T$. Does T_1 correspond to a rotation and translation about $\hat{x}_s$ and $\hat{y}_s$, respectively (a world-fixed transformation of T_{sa}), or a rotation and translation about $\hat{x}_a$ and $\hat{y}_a$, respectively (a body-fixed transformation of T_{sa})? Now calculate $T_2 = TT_{sa}$. Does T_2 correspond to a body-fixed or world-fixed transformation of T_{sa} ?
- (f) Use T_{sb} to change the representation of the point $p_b = (1, 2, 3)$ in $\{b\}$ coordinates to $\{s\}$ coordinates.
- (g) Choose a point p represented by $p_s = (1, 2, 3)$ in $\{s\}$ coordinates. Calculate $p' = T_{sb}p_s$ and $p'' = T_{sb}^{-1}p_s$. For each operation, should the result be interpreted as changing coordinates (from the $\{s\}$ frame to $\{b\}$) without moving the point p , or as moving the location of the point without changing the reference frame of the representation?
- (h) A twist $\mathcal{V}$ is represented in $\{s\}$ as $\mathcal{V}_s = (3, 2, 1, -1, -2, -3)$. What is its representation $\mathcal{V}_a$ in frame $\{a\}$?
- (i) By hand, calculate the matrix logarithm $[\mathcal{S}]\theta$ of T_{sa} . (You may verify your answer with software.) Extract the normalized screw axis $\mathcal{S}$ and rotation amount θ . Find a $\{q, \hat{s}, h\}$ representation of the screw axis. Redraw the fixed frame $\{s\}$ and in it draw $\mathcal{S}$.
- (j) Calculate the matrix exponential corresponding to the exponential coordinates of rigid-body motion $\mathcal{S}\theta = (0, 1, 2, 3, 0, 0)$. Draw the corresponding frame relative to $\{s\}$, as well as the screw axis $\mathcal{S}$.

4. (Exercise 3.17) Please solve the exercise discussed as follows.

Exercise 3.17 Four reference frames are shown in the robot workspace of Figure 3.23: the fixed frame $\{a\}$, the end-effector frame effector $\{b\}$, the camera frame $\{c\}$, and the workpiece frame $\{d\}$.

- (a) Find T_{ad} and T_{cd} in terms of the dimensions given in the figure.
- (b) Find T_{ab} given that

$$T_{bc} = \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

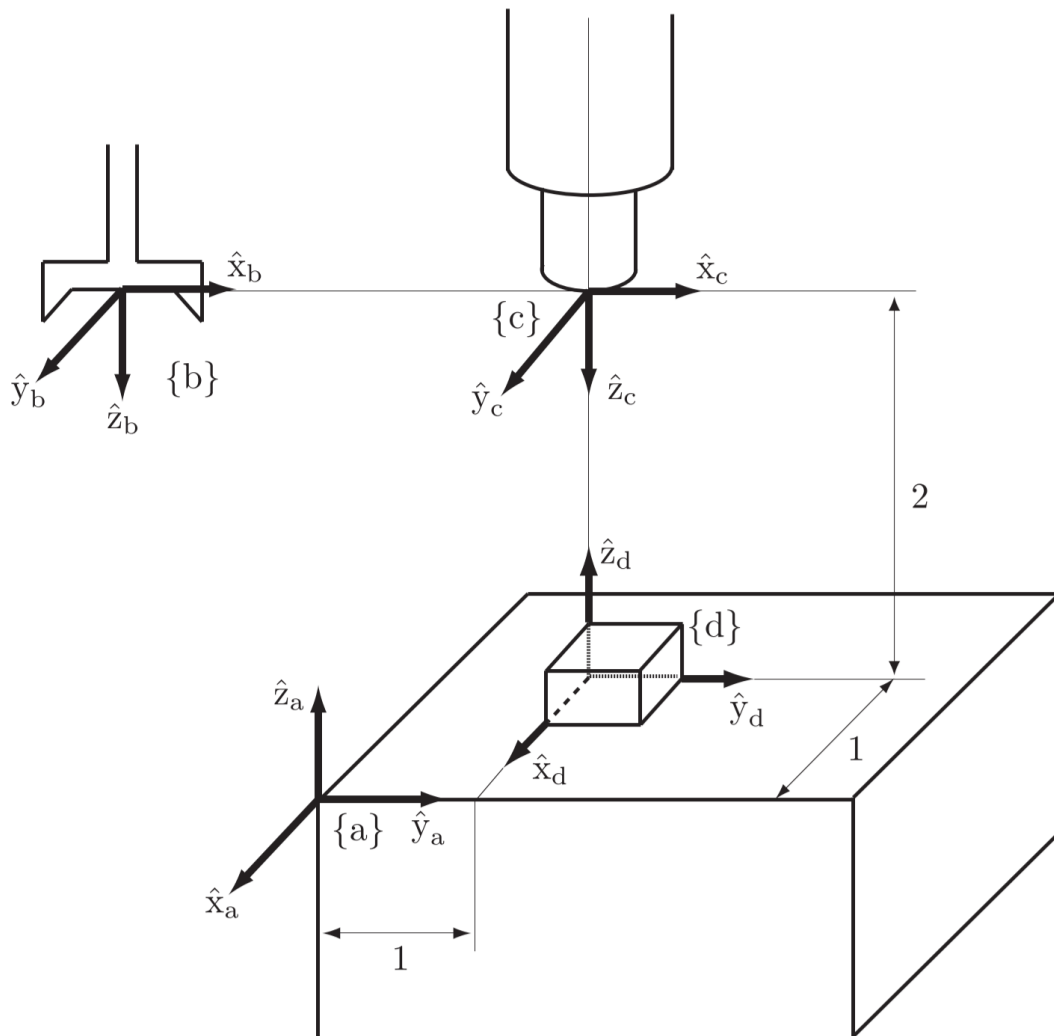


Figure 3.23: Four reference frames defined in a robot's workspace.

5. (Exercise 3.26) Find screw axis $\mathcal{S}$ provided that $q = (3, 0, 0)$, $\hat{s} = (0, 0, 1)$, and $h = 2$. Please draw the screw axis to verify your result.

6. (Exercise 3.27) Find velocity $\dot{\theta}$ and the screw axis $\mathcal{S}$ in terms of $\{q, \hat{s}, h\}$, provided the twist $\mathcal{V} = (0, 2, 2, 4, 0, 0)$. Please draw the screw axis to verify your result.